

MILKLOOP

Circular Milk Packaging

Design, Take-Back, Recovery &
Traceability | Responsible Business, Ethics
& Governance | PGDM – FIIB New Delhi

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The Problem

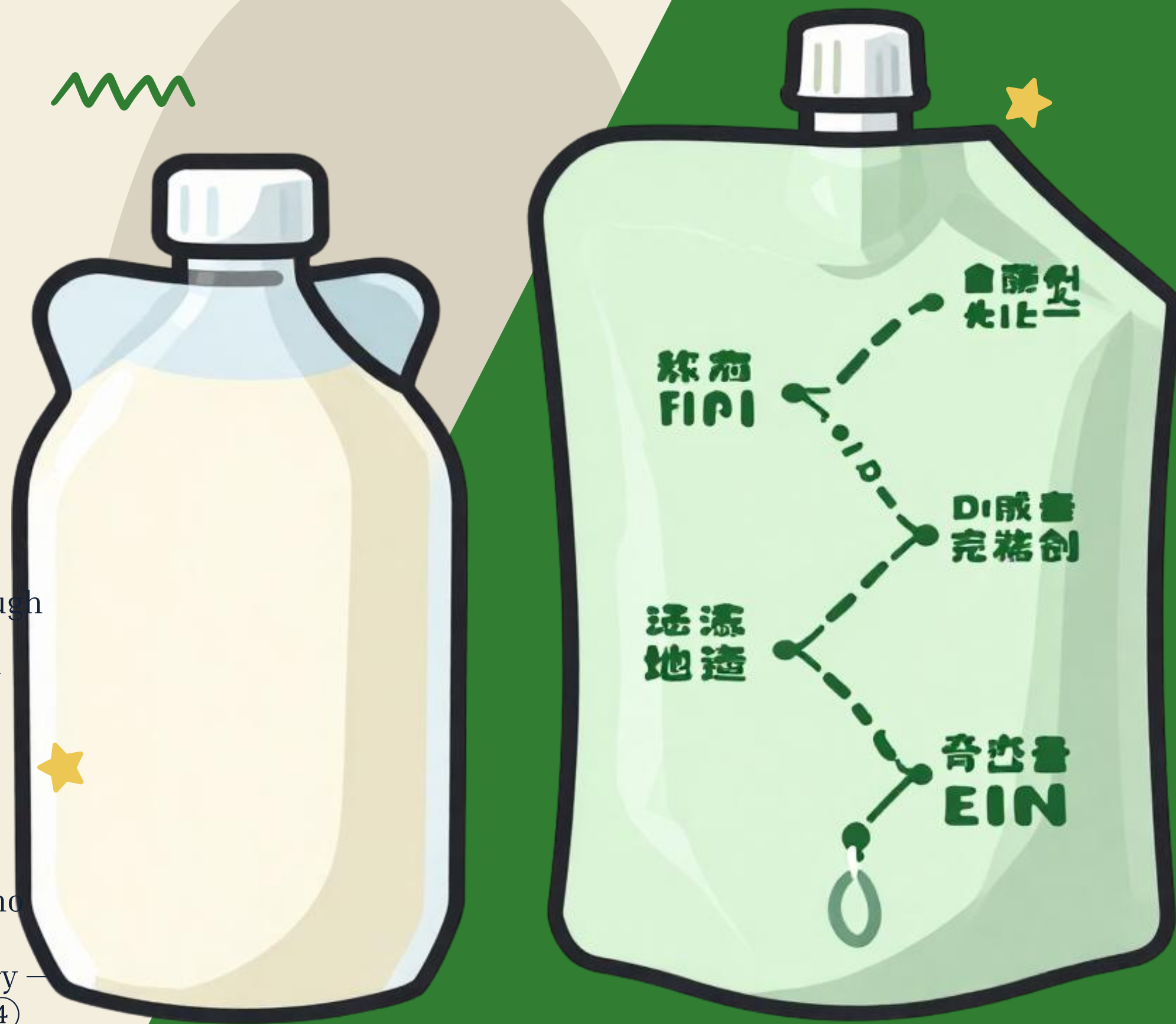
The Linear Milk-Pouch Problem

Take-Make-Dispose Journey

Raw materials are extracted and converted into LDPE/LLDPE film. Pouches are manufactured, filled at dairy plants, distributed through cold chains, and consumed by households — after which the used pouch enters a fragmented, uncontrolled post-consumer pathway with no guaranteed recovery or return.

Four Critical Breaks

① No controlled return pathway — pouches exit the system with no structured take-back. ② Material-value leakage — recoverable PE/LDPE is lost to landfill or open burning. ③ Fragmented recovery — informal collectors operate without coordination or traceability. ④ Evidence gap — no data confirms whether material is actually recovered or reused. Core insight: the pouch is not designed around its recovery journey.



Circular Packaging

From Disposable Pouch to Circular Packaging Asset

Conventional Model

Design → Produce → Fill → Distribute → Consumer Use → Dispose
Packaging is not designed around its recovery journey.
Material value is lost at end-of-use with no return pathway, no traceability, and no downstream productive re-entry.



MilkLoop Circular Model

Design → Milk Pouch → Consumer Use → Return → Recovery → Downstream Application → Productive Re-entry

Design-for-Circularity Principles:

- Mono-material PE/LDPE where technically suitable
- Optimise material use without compromising food safety
- Avoid incompatible components that hinder recovery
- Enable identification and sortability at recovery stage
- Design beyond point of sale — packaging as a recoverable asset



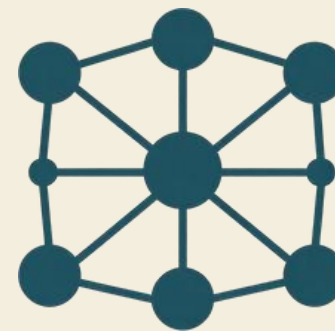
Closed-Loop Operating System

MilkLoop: Collect · Consolidate · Control · Circulate



Collect

Consumer returns used milk pouches via Kabadiwala, RWA collection point, Milk Booth, or MRF drop-off. Touchpoints are mapped, accessible, and integrated into daily routines to maximise participation and minimise friction at the source.



Coordinate & Control

MilkLoop Coordination Hub activates Digital Tracking, Local Aggregation, Sorting, Weighment & Quality Check, and Load Consolidation. Routing is optimised by actual volume and verified partner capacity. Collection ≠ Recycling ≠ Circularity.



Circulate

Verified Recycling/Recovery Partner receives consolidated loads. Recovered Material is quality-assured and directed to Appropriate Downstream Application, enabling Productive Re-entry and closing the loop with evidence-based traceability.

Reverse Logistics

Protecting the Margin

Optimised Collection & Routing



Cluster collection points at Kabadiwala hubs, RWAs and Milk Booths. Aggregate locally to build volume before movement. Consolidate loads to maximise truck utilisation. Route based on actual collection volume and verified partner capacity — not fixed schedules. Send only to certified recycling and recovery processors.

Unit Economics & Key Performance Indicators



Higher utilisation → lower logistics cost/kg → better contribution potential. KPIs: Collection Volume per Route | Logistics Cost/kg | On-Time Collection Rate | Contamination % | Recovery Yield | Material Re-Entry Rate. Economic gain is only valid when social and environmental standards are maintained.



Triple Bottom Line[★]

Does the Redesign Create Value?

People

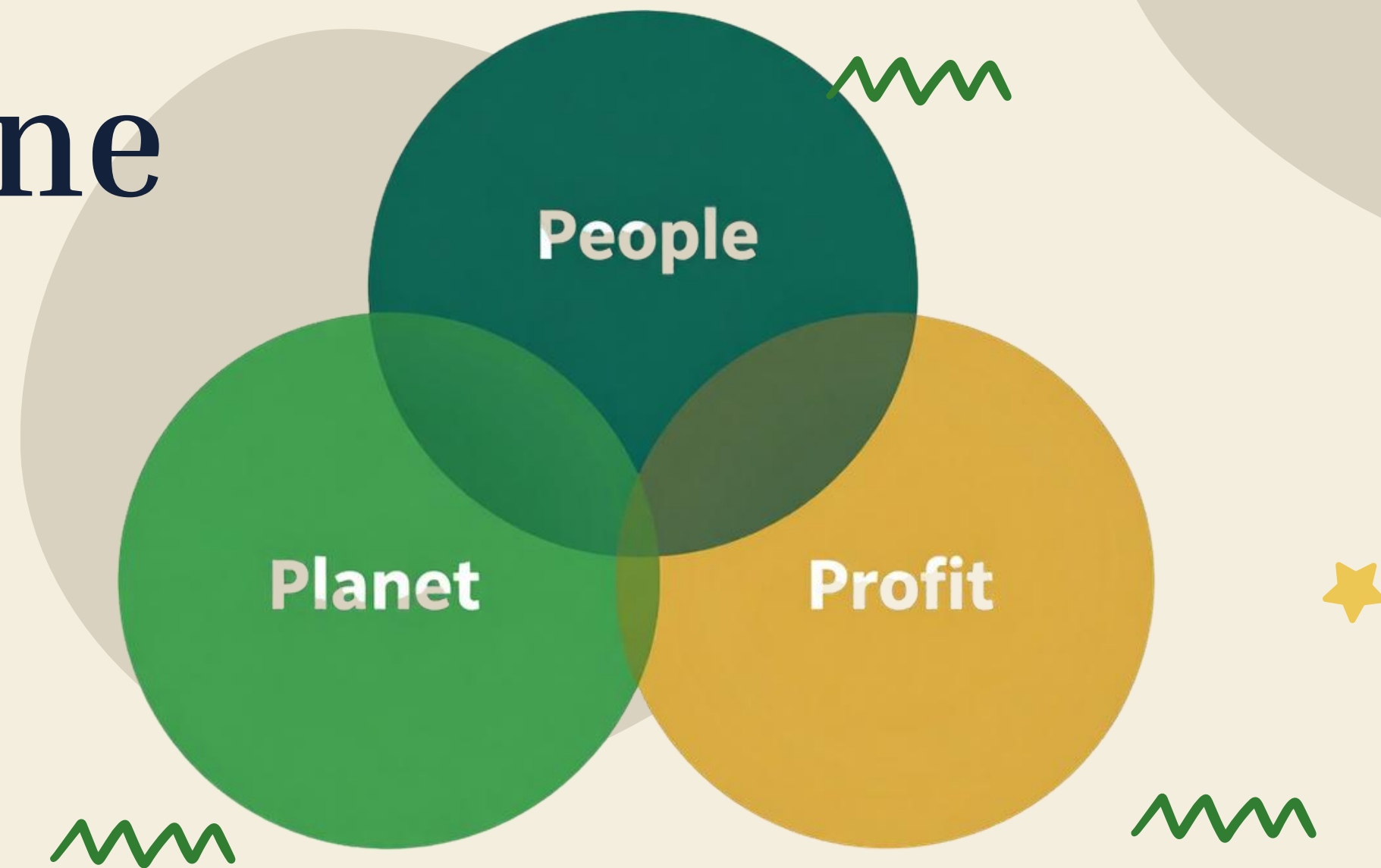
Organised collector participation with transparent, fair payments. Safety protocols and training for informal-sector workers. Inclusive stakeholder participation across the collection network. Dignity and visibility for Kabadiwalas and RWA partners in the recovery chain.

Planet

Systematic collection, processing and recovery of LDPE/LLDPE milk pouches. Verified recovered material with documented chain of custody. Productive re-entry into downstream applications. Rejection and residuals managed responsibly — no false circularity claims.

Profit

$\text{Revenue/kg} - \text{Operating Cost/kg} = \text{Contribution/kg}$. Gross margin tracked per collection route and recovery batch. Decision rule: economic gain is not success if it depends on unacceptable social or environmental harm. TBL integration is non-negotiable.



Financial Viability

TBL Trade-offs & Unit Economics

Revenue Streams & Stable Layers

Service revenue from recovery support forms the primary stable layer — traceability, reporting, and sustainability data provide recurring income less exposed to commodity volatility. Recovered-material value adds upside but remains sensitive to market price fluctuations. A diversified revenue base reduces dependence on any single stream.

Unit Economics & Break-Even

$\text{Revenue/kg} - \text{Operating Cost/kg} = \text{Contribution/kg}$. Break-even volume = $\text{Fixed Costs} \div \text{Contribution/kg}$. Broader collection increases volume but raises logistics cost/kg — the trade-off must be managed through route optimisation and load consolidation. No invented actuals; projections must be evidence-based.



Governance & Regulatory Architecture

Regulatory Framework

EPR Responsibility & MilkLoop's Role



Applicable EPR responsibility remains with the relevant dairy or brand owner. MilkLoop supports execution through four pillars: Coordinate • Track • Verify • Report. MilkLoop connects the collection network to verified processing partners and provides evidence-based reporting — it does not transfer statutory liability to any partner.

Governance Principles & Claim Discipline



Core governance principle: no sustainability claim is made without supporting evidence. MilkLoop enforces transparent data trails at every handoff — from collection to processing to verified recovery. Greenwashing is prevented through reconciled batch records and third-party verification at each stage of the circular loop.



How MilkLoop Proves Circularity

Weigh → Batch ID → Transfer → Process → Verify → Reconcile → Report



Traceability Chain

Every collection event is weighed and assigned a unique Batch ID. Material transfer is recorded at each handoff point — from collector to aggregator to verified processing partner — creating an unbroken, auditable chain of custody from source to recovery.



Verification Controls

Processing entry is confirmed, recovery yield is evidenced, and data is reconciled across the chain. Controls prevent false recovery claims, double counting, unsupported processing assertions, data manipulation, greenwashing, and worker exploitation at any node.



Claim Discipline

Collected = material received and weighed. Processed = entered verified processing. Recovered = usable output evidenced by processor. Circular = downstream productive use demonstrated. No claim is made without supporting evidence at each stage.

Stakeholder Value



Why Would Any Stakeholder Participate?

Dairy Companies & Collectors

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Dairy Companies: Organised recovery support, end-to-end traceability, reporting-ready evidence for EPR compliance, and verified circularity claims. Kabadiwalas/Collectors: Predictable demand, assured buyer, transparent and timely payments, organised collection arrangements, and improved working conditions.



RWAs, MRFs & Recyclers

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RWAs: Convenient, low-effort collection infrastructure within residential communities. MRFs: Additional material flow and organised downstream linkage. Recyclers/Processors: Predictable, quality-sorted feedstock with full traceability from source to processing facility.

Consumers & Feedback Loop

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Consumers: Low-friction return mechanism with visible, measurable environmental impact. Two-Way Engagement: All stakeholders connected through a structured feedback, grievance, and remediation loop – ensuring transparency, accountability, and continuous improvement across the MilkLoop ecosystem.

Conclusion

Creating Shared Value: Why MilkLoop Is a Business

The Value Creation Chain

Plastic-waste problem → Stakeholder participation → Organised material recovery → Verified circular material → Two value streams: Circular packaging services + Recovered-material value → Recurring commercial relationships built on People, Planet, Profit & Governance trust.

The MilkLoop Proposition

Packaging redesigned for circularity. Supply chain redesigned for return. Business model monetises coordination and evidence. MilkLoop is not just a recycling initiative — it is a commercially viable circular system.

